| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 10/13/2017 |
| **PI Name:** | Axelbaum, Richard Ph.D. | | |
| **Project Title:** | Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Combustion science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | rla@wustl.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 314 935-7560 |
| **Organization Name:** | Washington University | | |
| **PI Address 1:** | Department of Energy, Environmental and Chemical Engineering | | |
| **PI Address 2:** | 305 Jolley Hall | | |
| **PI Web Page:** | | | |
| **City:** | St. Louis | **State:** | MO |
| **Zip Code:** | 63130 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 97-HEDS-01 |
| **Start Date:** | 12/12/2014 | **End Date:** | 12/11/2019 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Stocker, Dennis P | **Contact Phone:** | 216-433-2166 |
| **Contact Email:** | dennis.p.stocker@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17) | | |
| **Key Personnel Changes/Previous PI:** | October 2017 report: Beei-Huan Chao, University of Hawaii, has retired and resigned from his role as Flame Design Co-Investigator. | | |
| **COI Name (Institution):** | Sunderland, Peter Ph.D. ( University of Maryland )<br>Urban, David Ph.D. ( NASA Glenn Research Center ) | | |
| **Grant/Contract No.:** | NNX15AC75A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A).<br>The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants.<br><br>This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants. |
| **Task Progress:** | The laser-induced breakdown spectroscopy (LIBS) measurement method has been improved at Washington University in St. Louis. This includes a new oscilloscope setup and preparation for better acoustic monitoring. We have synchronized the new oscilloscope with the LIBS system. We completed work on noise filtering of the acoustic measurements. We performed Matlab programming on automated acoustic aided LIBS measurements. We performed high temperature LIBS tests in a Hencken burner, LIBS tests in a McKenna burner under same equivalence ratio and different temperature, and acoustic aided LIBS measurements in a McKenna burner with varying equivalence ratio. We adapted a maximum likelihood estimation method instead of average in LIBS data processing; compared and correlated the baseline for different element emission lines; and worked on the scalar measurement. We investigated the compositional effects in laser-induced backward transfer (LIBT) measurement and investigated the screening methods of LIBT measurement by looking into the statistics. We investigated the simultaneous composition and temperature measurement using laser induced plasma. We explored the water-gas-shift equilibrium in higher hydrocarbon flames and investigated the potential of inferring molecular profiles from LIBS measured elemental compositions. We investigated the water-gas-shift equilibrium in higher hydrocarbons using detailed mechanisms for heptane and toluene. We developed a plasma model to simulate the sound wave generation as a function of laser energy input and local gas properties and to optimize the measurements using a CCD based spectrometer.<br>Wendong Wu successfully defended his Ph.D. in May, 2017, at Washington University in St. Louis. The collaboration between Washington University and Shanghai Jiaotong University will continue in the field of laser plasma diagnostics. Graduate student Pan Du has joined the Flame Design team.<br><br>At the University of Maryland, we continue to examine the effects of dilution on lengths of normal and inverse gas jet diffusion flames. The method of measuring flame lengths was refined to examine only the blue intensities in the recorded images. Both normal and inverse flames now correlate together according to the theory of Roper. We have also characterized the double flame structure visible in some inverse diffusion flames. We imaged the double flames at 310 and 430 nm. We are measuring oxygen concentrations and temperatures in and above the double flames. This will help to determine if oxygen in this region contributes to the formation of the double luminous zones. A paramagnetic oxygen analyzer is being used, which has a mole fraction range of 0-1 and an accuracy of 0.001%.<br><br>At the University of Hawaii Manoa, Beei-Huan Chao has retired and resigned from his role as Flame Design Co-Investigator. His invaluable contributions to Flame Design are greatly appreciated and will be missed going forward. Prior to his retirement, progress was made on analytical studies of C/O ratio in diffusion flames. We examined extinction of spherical diffusion flames attached to a burner and revisited reactant leakage of diffusion flames. We revisited Linan's 1974 paper following the “total enthalpy” concept.<br><br>The research team published two journal papers, four conference papers and abstracts, and one Ph.D. dissertation.<br><br>The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 2 s drop facility. |
| **Bibliography Type:** | Description: (Last Updated: 11/26/2025) |
| **Abstracts for Journals and Proceedings** | Wang Z, Sunderland PB, Axelbaum RL. "Images of Co-Axial Gas Jet Diffusion Flames." Art Competition Poster. 2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017.<br>2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017. , Apr-2017 |
| **Abstracts for Journals and Proceedings** | Wang Z, Wu W, Sunderland PB, Axelbaum RL. "Ground-Based Research Supporting Flame Design." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017.<br>33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017 |
| **Articles in Peer-reviewed Journals** | Rodenhurst MK 3rd, Chao B-H, Sunderland PB, Axelbaum RL. "Structure and extinction of spherical burner-stabilized diffusion flames that are attached to the burner surface." Combustion and Flame. 2018 Jan;187:22-9. Available online 21 September 2017. https://doi.org/10.1016/j.combustflame.2017.08.024 (copy and paste URL for valid connectivity) , Jan-2018 |

| | |
|---|---|
| **Articles in Peer-reviewed Journals** | Gopan A, Yang Z, Kumfer BM, Axelabaum RL. "Effects of inert-placement (Zst) on soot and radiative heat flux in turbulent diffusion flames." Energy and Fuels. 2017 Jul 20;31(7):7617-23. https://doi.org/10.1021/acs.energyfuels.7b01092 , Jul-2017 |
| **Dissertations and Theses** | Wu W (Wendong Wu). "Fundamental Studies of Flame Structure Through Laser Plasma Diagnostics." Ph.D. Dissertation, Washington University in St. Louis, May, 2017. , May-2017 |
| **Dissertations and Theses** | El Makawy MH (Mohamed H. El Makawy). "The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames." M.S. Thesis, Ghent Univ., Department of Flow, Heat and Combustion, 54 p., Aug. 2016. , Aug-2016 |
| **Papers from Meeting Proceedings** | Wang Z, Sunderland PB, Axelbaum RL. "Observations of Double Flames in Inverse Gas Jet Diffusion Flames." 2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017.<br>2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017. 6 p. , Apr-2017 |
| **Papers from Meeting Proceedings** | Wang Z, Sunderland PB, Axelbaum RL "The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames." 2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017.<br>2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017. 6 pp. , Apr-2017 |